

No. of Functional Features Included in the Solution

Date	15 July 2026
Project Name	Smart Lender – AI Powered Loan Approval Prediction System
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	Loan Eligibility Prediction	Predicts whether a loan application is approved or rejected using Machine Learning.	Prediction Module	Done	High
2	Data Preprocessing	Cleans and transforms applicant data before model training.	Data Processing	Done	High
3	Machine Learning Model Training	Trains Decision Tree, Random Forest, KNN, and XGBoost models.	ML Module	Done	High
4	Model Evaluation	Compares model performance and selects the best-performing model.	Evaluation Module	Done	In Medium
5	Web-Based User Interface	Allows users to enter loan details and receive instant predictions.	Flask Web Application	Done	High
6	Real-Time Prediction	Generates instant loan approval predictions using the trained model.	Prediction Engine	Done	High
7	IBM Cloud Deployment	Supports deployment of the application on IBM Cloud.	Deployment Module	In Progress	In Medium
8	Input Validation & Error Handling	Validates user inputs and handles invalid data gracefully.	Validation Module	Done	In Medium

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	6
Additional / Nice-to-Have Features	2
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	2	Loan Application Form, Prediction Result Page
2	Backend / Logic	3	Data Preprocessing, Model Training, Loan Prediction
3	Database / Storage	1	Dataset Storage and Model Files
4	API / Integration	1	Flask Application Integration
5	Security / Authentication	1	Input Validation and Error Handling